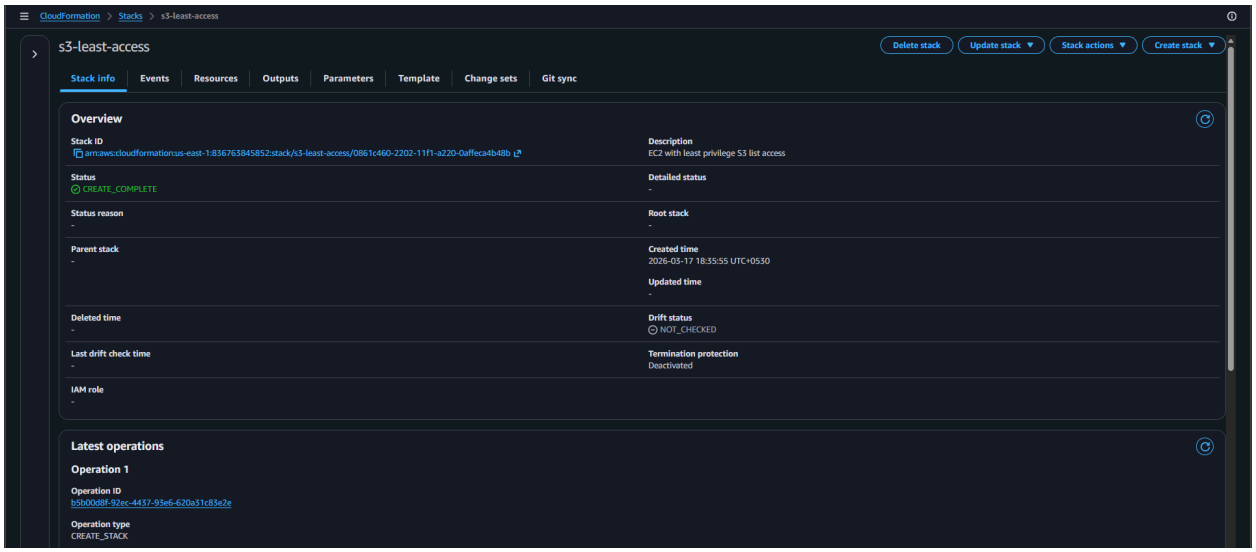
S3 least Privilege access

Output

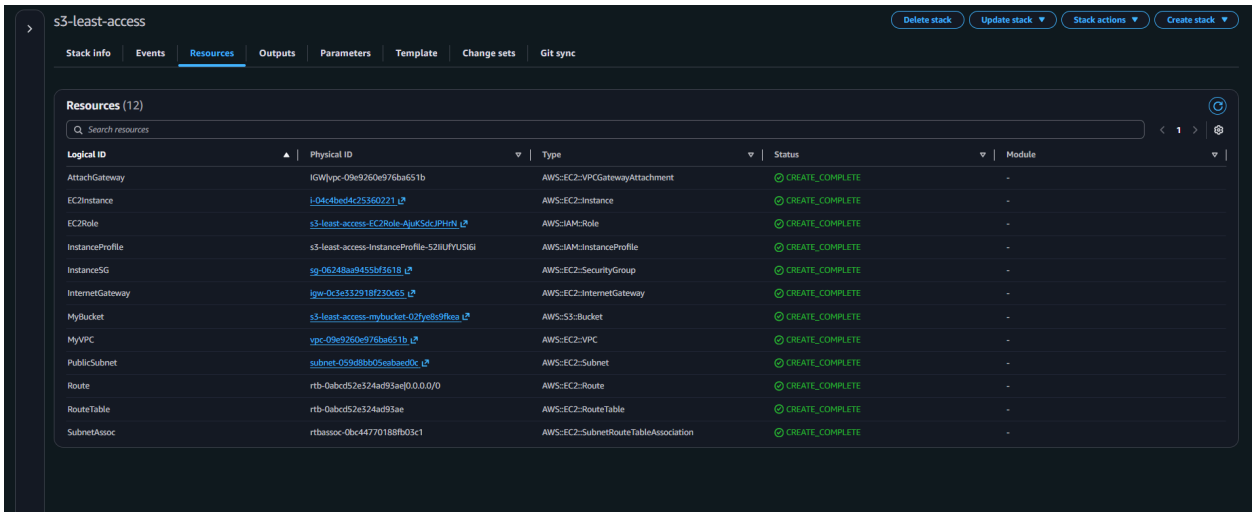
```
[ec2-user@ip-10-0-1-84 ~]$ aws s3 ls s3://s3-least-access-mybucket-02fye8s9fkea
[ec2-user@ip-10-0-1-84 ~]$ aws s3 ls s3://s3-least-access-mybucket-02fye8s9fkea
2026-03-17 13:22:46    5275297 cat.png
[ec2-user@ip-10-0-1-84 ~]$ aws s3 ls

An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:sts::836763845852:assumed-r
ole/s3-least-access-EC2Role-AjuKSdcJPHrN/i-04c4bed4c25360221 is not authorized to perform: s3:ListAllMyBuckets bec
ause no identity-based policy allows the s3:ListAllMyBuckets action
[ec2-user@ip-10-0-1-84 ~]$
```

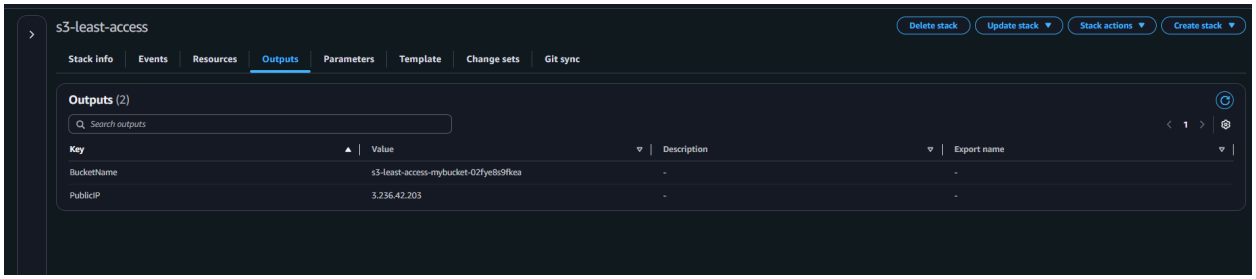
Stack info



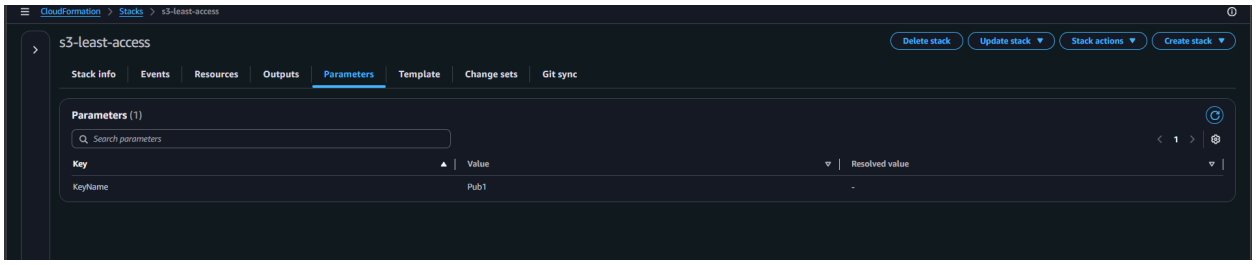
Resources



Output



Parameters



Template

```yaml
AWSTemplateFormatVersion: "2010-09-09"
Description: EC2 with least privilege S3 list access

Parameters:
  KeyName:
    Type: AWS::EC2::KeyPair::KeyName

Resources:

# ---------------- VPC ----------------
  MyVPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true

# ---------------- Subnet ----------------
  PublicSubnet:
    Type: AWS::EC2::Subnet
    Properties:
      VpcId: !Ref MyVPC
      CidrBlock: 10.0.1.0/24
      MapPublicIpOnLaunch: true

# ---------------- IGW ----------------
  InternetGateway:
    Type: AWS::EC2::InternetGateway

  AttachGateway:
    Type: AWS::EC2::VPCGatewayAttachment
    Properties:
      VpcId: !Ref MyVPC
      InternetGatewayId: !Ref InternetGateway

# ---------------- Route Table ----------------
  RouteTable:
    Type: AWS::EC2::RouteTable
    Properties:
      VpcId: !Ref MyVPC

  Route:
```

```yaml
    Type: AWS::EC2::Route
    DependsOn: AttachGateway
    Properties:
      RouteTableId: !Ref RouteTable
      DestinationCidrBlock: 0.0.0.0/0
      GatewayId: !Ref InternetGateway

  SubnetAssoc:
    Type: AWS::EC2::SubnetRouteTableAssociation
    Properties:
      SubnetId: !Ref PublicSubnet
      RouteTableId: !Ref RouteTable

# ---------------- SG ----------------
  InstanceSG:
    Type: AWS::EC2::SecurityGroup
    Properties:
      GroupDescription: Allow SSH
      VpcId: !Ref MyVPC
      SecurityGroupIngress:
        - IpProtocol: tcp
          FromPort: 22
          ToPort: 22
          CidrIp: 0.0.0.0/0

# ---------------- S3 ----------------
  MyBucket:
    Type: AWS::S3::Bucket

# ---------------- IAM Role ----------------
  EC2Role:
    Type: AWS::IAM::Role
    Properties:
      AssumeRolePolicyDocument:
        Version: "2012-10-17"
        Statement:
          - Effect: Allow
            Principal:
              Service: ec2.amazonaws.com
            Action: sts:AssumeRole
      Policies:
        - PolicyName: S3ListOnly
          PolicyDocument:
            Version: "2012-10-17"
```

```yaml
          Statement:
            - Effect: Allow
              Action:
                - s3:ListBucket
              Resource: !GetAtt MyBucket.Arn

  InstanceProfile:
    Type: AWS::IAM::InstanceProfile
    Properties:
      Roles:
        - !Ref EC2Role

# ---------------- EC2 ----------------
  EC2Instance:
    Type: AWS::EC2::Instance
    Properties:
      InstanceType: t3.micro
      KeyName: !Ref KeyName
      ImageId: ami-02dfbd4ff395f2a1b
      IamInstanceProfile: !Ref InstanceProfile

      NetworkInterfaces:
        - DeviceIndex: 0
          SubnetId: !Ref PublicSubnet
          AssociatePublicIpAddress: true
          GroupSet:
            - !Ref InstanceSG

      UserData:
        Fn::Base64: !Sub |
          #!/bin/bash
          yum install -y aws-cli
          aws s3 ls s3://${MyBucket} > /home/ec2-user/output.txt

# ---------------- Outputs ----------------
Outputs:
  PublicIP:
    Value: !GetAtt EC2Instance.PublicIp

  BucketName:
    Value: !Ref MyBucket
```